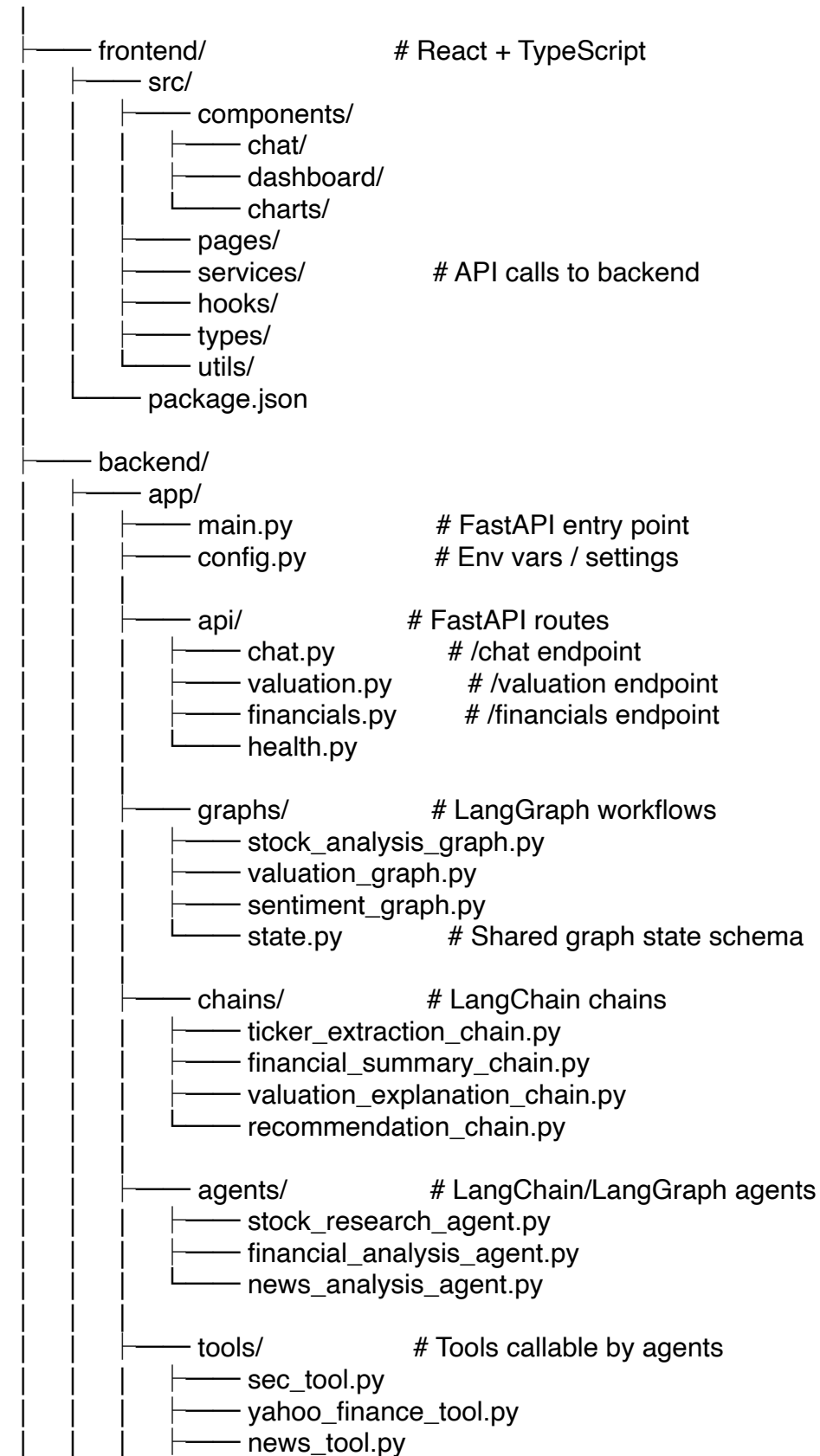


ai-stock-valuation-chatbot/



```
├── ratio_tool.py
├── dcf_tool.py
├── sentiment_tool.py
├── macro_tool.py
├── services/          # Non-LLM business logic
│   ├── financial_data_service.py
│   ├── ratio_service.py
│   ├── vertical_analysis_service.py
│   ├── horizontal_analysis_service.py
│   ├── dcf_service.py
│   ├── sentiment_service.py
│   └── recommendation_service.py
├── prompts/          # Prompt templates
│   ├── system_prompts.py
│   ├── valuation_prompt.py
│   ├── financial_summary_prompt.py
│   └── recommendation_prompt.py
├── memory/           # Chat/session memory
│   ├── conversation_memory.py
│   └── checkpointing.py  # LangGraph checkpoints
├── db/
│   ├── database.py
│   ├── models.py
│   ├── schemas.py
│   └── repositories/
│       ├── company_repository.py
│       ├── financials_repository.py
│       └── news_repository.py
├── data_ingestion/
│   ├── sec_ingestion.py
│   ├── yahoo_ingestion.py
│   ├── news_ingestion.py
│   └── macro_ingestion.py
├── utils/
│   ├── ticker_utils.py
│   ├── formatting.py
│   └── validation.py
├── tests/
└── requirements.txt
```

